Revision 2

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- Matrices
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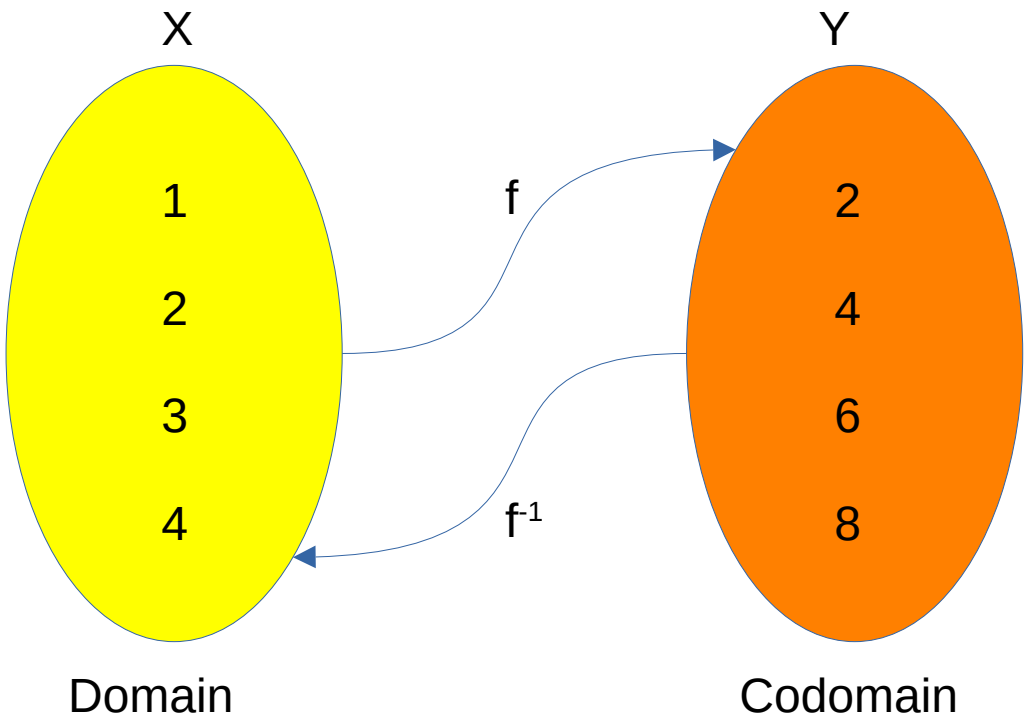
  - Properties of Matrices
  - Eigenvectors and Eigenvalues
  - Row Echelon Form and Reduced Row Echelon Form
  - Elementary Row Operations and Gaussian Elimination
  - Gradients And Hessians

Functions

Functions

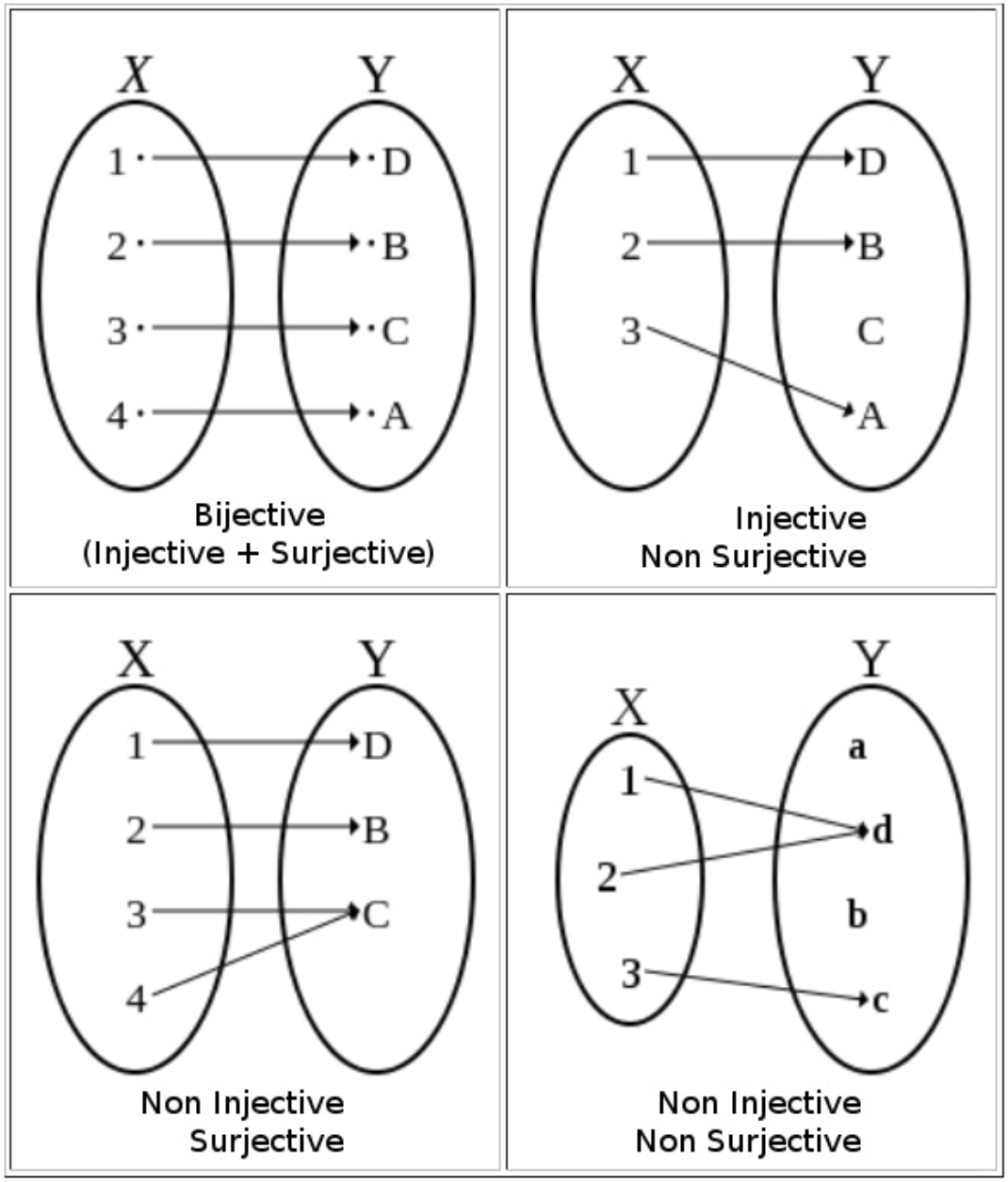
- A mapping in certain form, for example: f: $\mathbb{R} \rightarrow \mathbb{R}$
- Example: $f(x) = 2x$, $g(x) = x^2 - 1$
- Each element in domain can be mapped to exactly 1 element in codomain.



Injective and Surjective

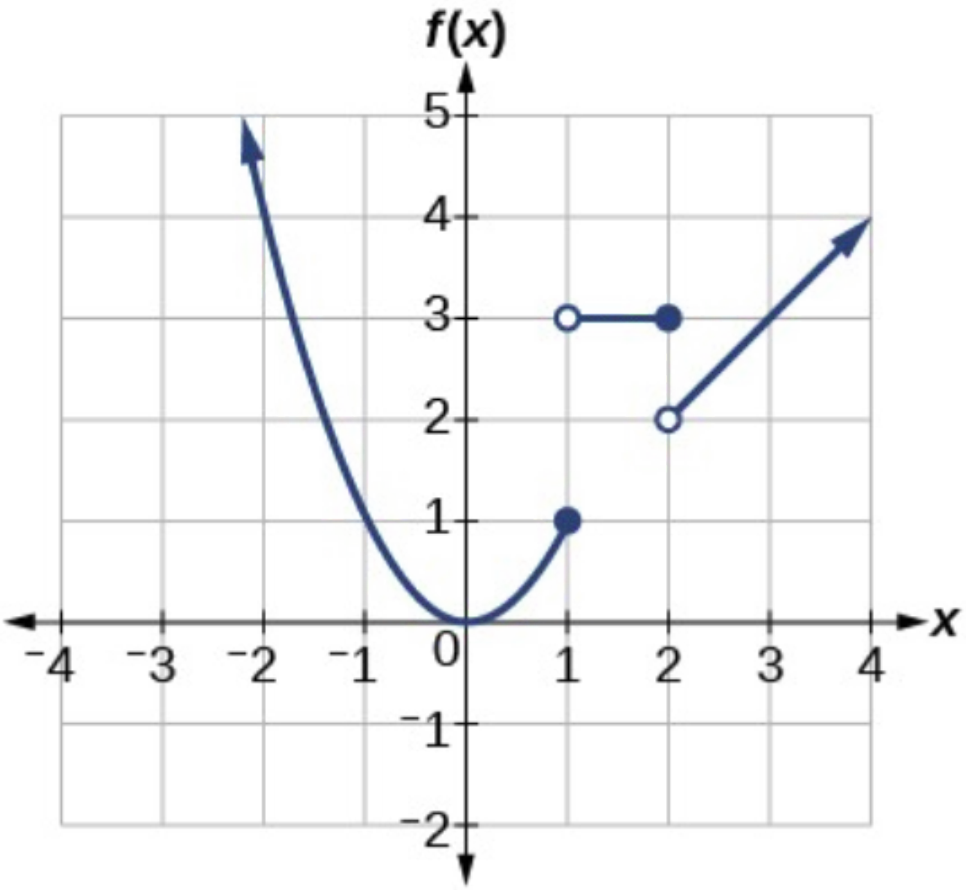
- Injective / One-to-one functions: for each $y \in Y$, there is at most one $x \in X$ with $f(x)=y$
- Surjective/onto functions: the range of f is Y (codomain)
- Bijective functions: functions which are both injective and surjective

Example


X
Y
1
2
3
4
D
B
C
A
Bijective
(Injective + Surjective)
X
Y
1
2
3
D
B
C
A
Injective
Non Surjective
X
Y
1
2
3
4
D
B
C
Non Injective
Surjective
X
Y
1
2
3
a
d
b
c
Non Injective
Non Surjective

Piecewise Functions

A function that behaves differently based on input x is called a piecewise function.



Example

$$f(x) = \begin{cases} x^2, & if\ x \leq 1 \\ 3, & if\ 1 < x \leq 2 \\ x, & if\ x > 2 \end{cases}$$

Multivariable Functions

- Some functions have more than one variables.

  f: X x Y → Z g: $\mathbb{R}^n \to \mathbb{R}$

- where X, Y and $\mathbb{R}^n$ are domains; Z and $\mathbb{R}$ are codomains

- Sometimes, we can consider the variables of these as vectors.

- Examples:

  h(x,y) = y-x

  h(10,3) = 10-3=7

$$f(x,y) = \begin{cases} \frac{y}{x} - y & \text{if} \quad 0 \le y < x \le 1 \\ \frac{x}{y} - x & \text{if} \quad 0 \le x < y \le 1 \\ 1 - x & \text{if} \quad 0 < x = y \\ 0 & \text{everywhere else.} \end{cases}$$

Example: Multivariable piecewise function

Matrix

Matrix

- A rectangular array of values

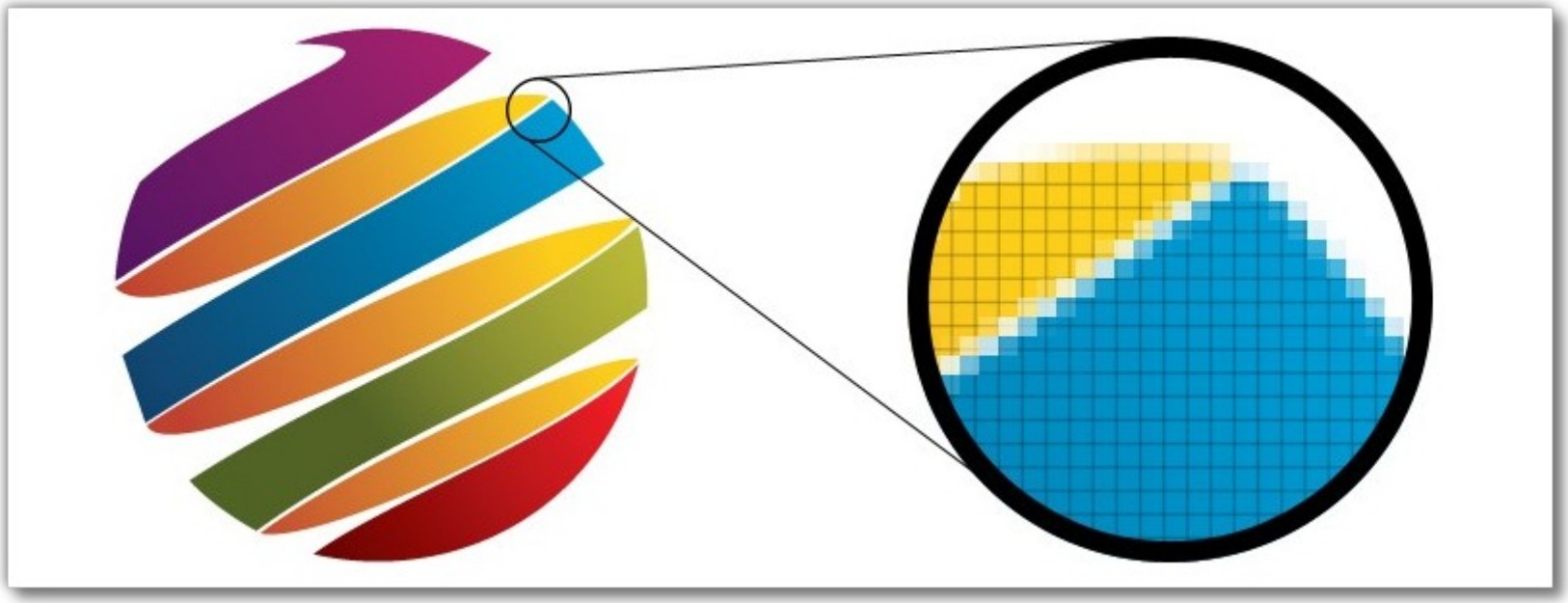
$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}}_{n \text{ columns}} \left.\vphantom{\begin{bmatrix} a \\ a \\ a \\ \vdots \\ a \end{bmatrix}}\right\} m \text{ rows}$$

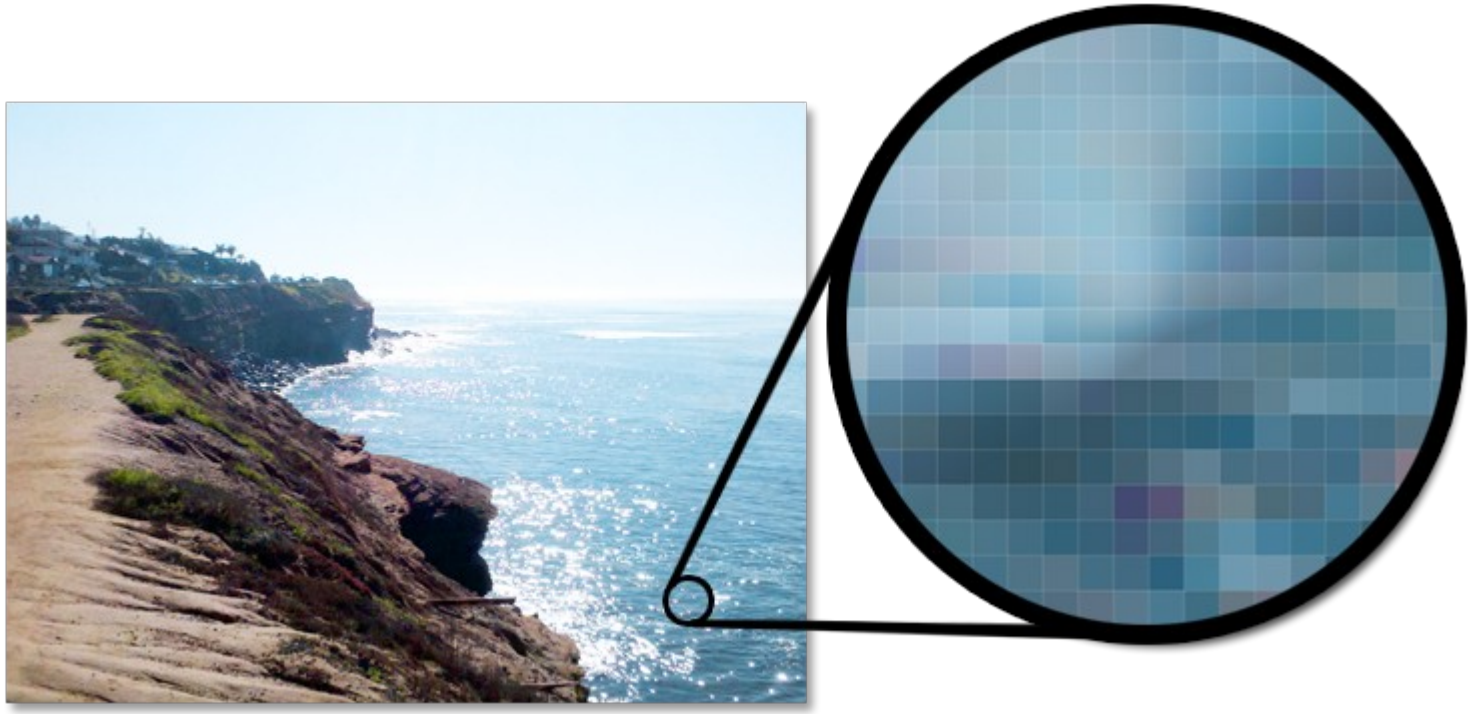
Each entry, a_{ij} refers to the value in row i, column j.

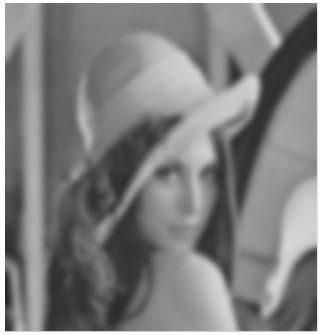
If all entries for A are real numbers, $A \in \mathbb{R}^{mxn}$

Why Matrix is so important?

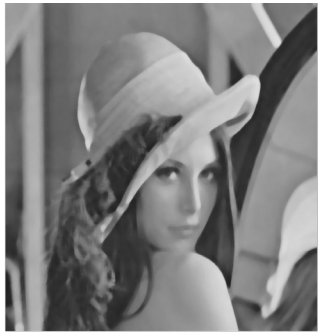
- Represent data in rectangular grid form (e.g. datasets in machine learning)
- To solve systems of linear equations (imagine you have systems of hundreds, or even thousands of equations/variables)
- Image / video processing
- Represent graphs







Blur



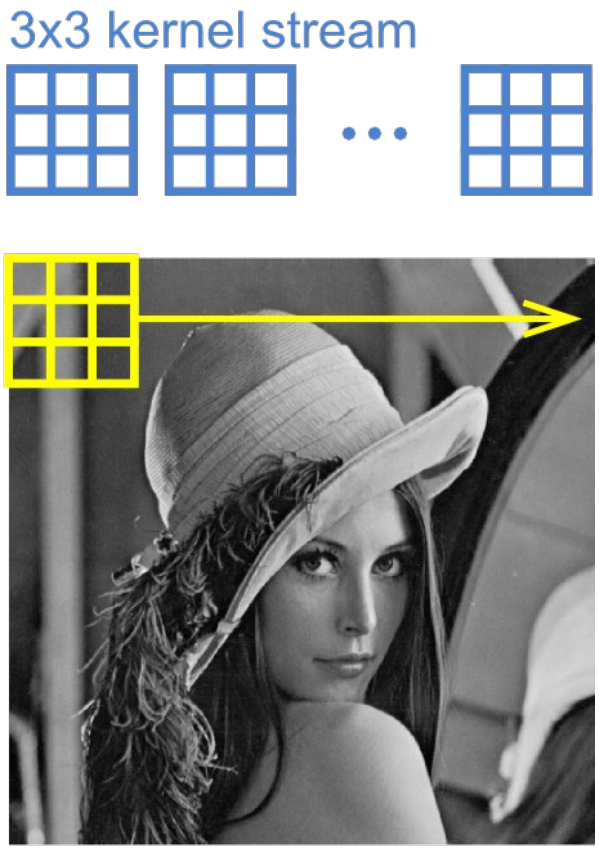
Median

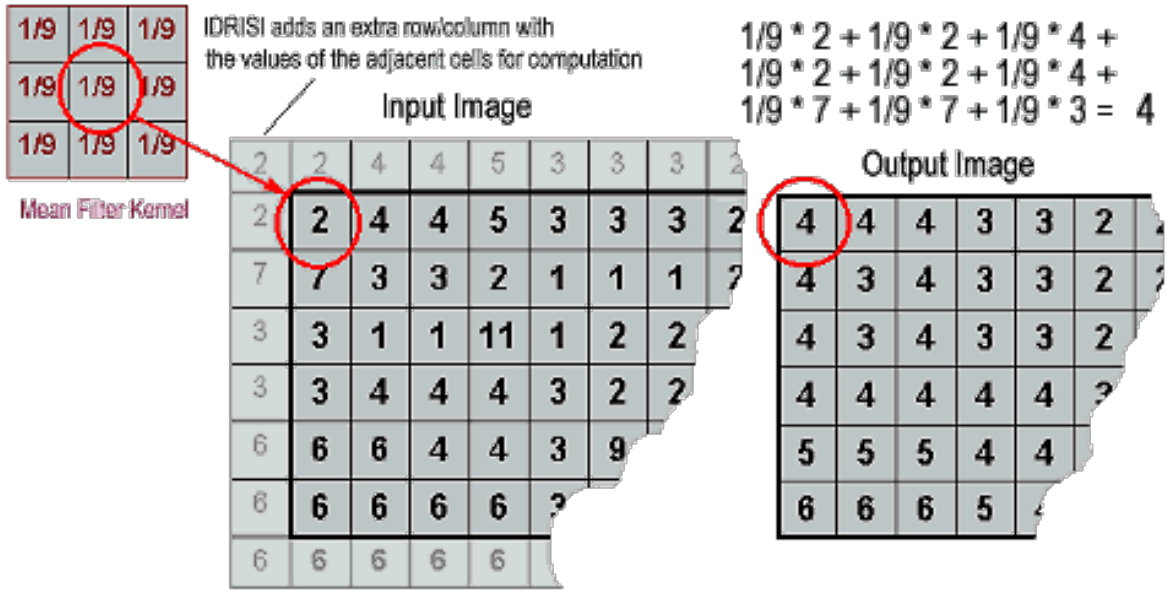
Edge-Detect

High-Pass

Dilate

Erode





In mathematics, matrices are studied within the domain of linear algebra.

Addition and Subtraction

We can add/subtract two matrices (of the same dimension) by adding their corresponding entries.

$$\begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 3 \\ -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 + 1 & 2 + 3 \\ 0 + (-1) & 1 + 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 5 \\ -1 & 3 \end{bmatrix}$$

Scalar Multiplication

- We can multiply a matrix A by a scalar c by multiplying each entry in A by c

Scalar Matrix

$$3\begin{bmatrix} -1 & 2 \\ 6 & 5 \end{bmatrix} = \begin{bmatrix} 3(-1) & 3(2) \\ 3(6) & 3(5) \end{bmatrix} = \begin{bmatrix} -3 & 6 \\ 18 & 15 \end{bmatrix}$$

Matrix Multiplication

- A matrix multiplication is a product of 2 matrices.

Requirement for *AB*:

size of A is m x n and

size of B is n x p

Result for *AB*:

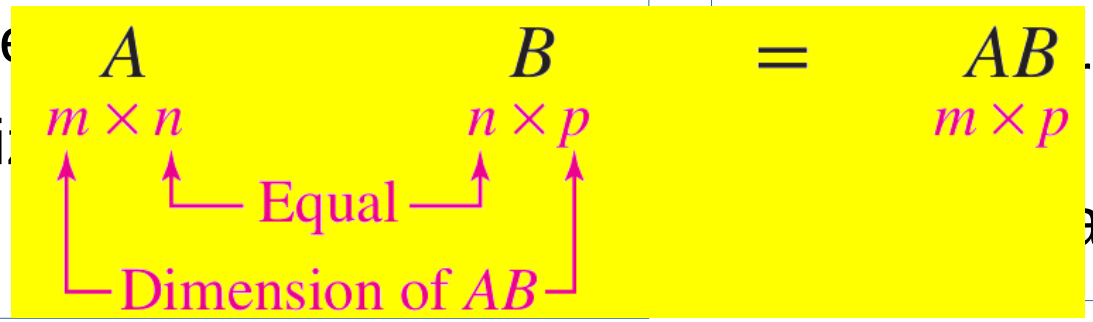
m x p matrix

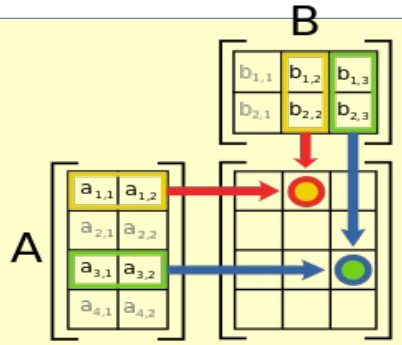
If $A = [a_{ij}]$ and $B = [b_{ij}]$

$AB = [c_{ij}]$ where $c_{ij} = a_{i1}\, b_{1j} + a_{i2}\, b_{2j} + a_{i3}\, b_{3j} + \ldots + a_{in}\, b_{nj}$

Matrix Multiplication

- A matrix multiplication is a product of 2 matrices.





Types of Matrices

Square Matrix

- a square matrix is a n x n matrix where n is a positive Integer.

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Identity Matrix

- An identity matrix, I is a square matrix such that all diagonal entries are 1s, all other entries are 0s.
- AI = A
- I_n is diagonal matrix with dimension n.
- Examples: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

  Examples of non-identity matrix:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Diagonal Matrix

- All the entries not at the diagonal are 0.
- Identity matrix is a special case of diagonal matrix

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{bmatrix} \qquad \begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \end{bmatrix}$$

Symmetric matrix

- Must be a square matrix,
- With symmetric property

$$\begin{bmatrix} 3 & 7 & -8 \\ 7 & -1 & 2 \\ -8 & 2 & 4 \end{bmatrix}$$

Upper / Lower Triangular Matrix

- Upper triangular matrix, U is defined as:

$$U_{ij} = \begin{cases} a_{ij} & \text{for } i \leq j \\ 0 & \text{for } i > j \end{cases}$$

$$\begin{bmatrix} 3 & 7 & -8 & 8 \\ 0 & -1 & 6 & 2 \\ 0 & 0 & 4 & 12 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

- Lower triangular matrix, U is defined as:

$$U_{ij} = \begin{cases} 0 & \text{for } i < j \\ a_{ij} & \text{for } i \geq j \end{cases}$$

$$\begin{bmatrix} 3 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ -6 & -4 & 4 & 0 \\ 3 & -5 & 9 & 5 \end{bmatrix}$$

Column and Row Vector

- When row=1, we have row vector

$$\begin{bmatrix} 1 & 3 & 5 & 7 \end{bmatrix}$$

- When column=1, we have column vector

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

Properties of Matrices

Transpose of Matrix

- If A is a m x n matrix, transpose of A, or A^T is a n x m matrix obtained from A by interchanging the rows and columns of A.
- After transpose,
  - Column i of A $\rightarrow$ Row i of A^T
  - Row j of A $\rightarrow$ Column j of A^T

$$AB \neq BA$$

$$(AB)^T = B^T A^T$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \qquad A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

Revisit Column and Row Vector

- When row=1, we have row vector

$$\begin{bmatrix} 1 & 3 & 5 & 7 \end{bmatrix}$$

- When column=1, we have column vector

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

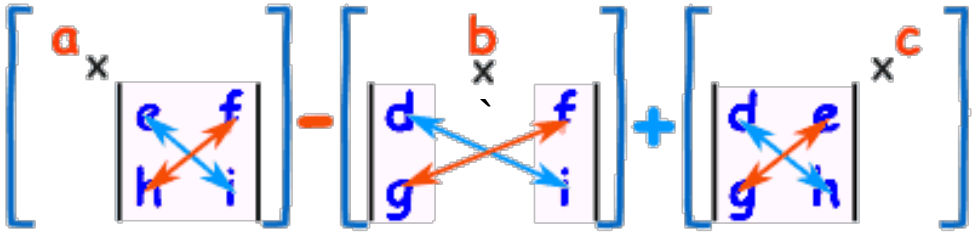
Note: if $x \in \mathbb{R}^n$ $A \in \mathbb{R}^{n \times n}$

$x^T x \in \mathbb{R}$ $xx^T \in \mathbb{R}^{n \times n}$

$x^T A x \in \mathbb{R}$

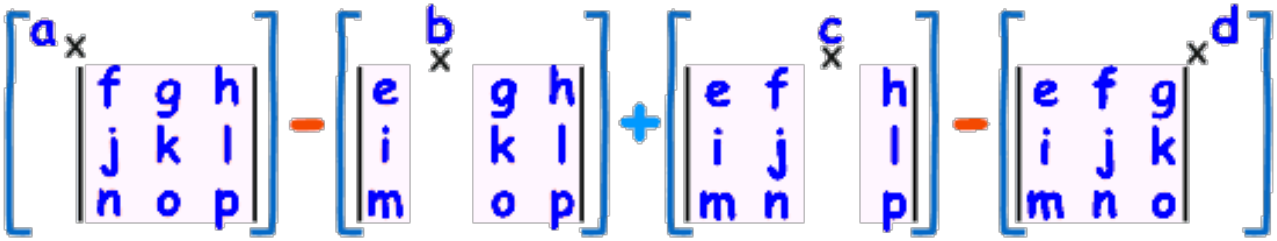
Determinant of Matrices

- Scalar value that characterizes properties of square matrices
- If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, det(A) = ad – bc
- If $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$,



$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

4X4 matrices:



Inverse of a matrix

- Let A and B are square matrices with dimensions n and In is the identity matrix.
- If $AB = BA = I_n$

  then B is the inverse of A.
- We can write B as A^{-1}
- $(A^{-1})^{-1}=A$
- $(AB)^{-1}=B^{-1}A^{-1}$

Inverse of a matrix

- Let A and B are square matrices with dimensions n and In is the identity matrix.
- If $AB = BA = I_n$

  then B is the inverse of A.
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- $(A^{-1})^{-1}=A$
- $(AB)^{-1}=B^{-1}A^{-1}$

If the determinant of a matrix is 0, the matrix has no inverse.

A matrix with no inverse is a singular matrix. Otherwise, it is nonsingular.

Eigenvector and Eigenvalue

What is Eigenvector and Eigenvalue

- When a square matrix A multiply with a nonzero vector x, it changes the direction of x, except when x is the eigenvector of A.

$$Ax = \lambda x$$

- A : nxn matrix
- x : eigenvector (non zero)
- λ : eigenvalue
-

Finding Eigenvector and Eigenvalue

$$Ax = \lambda x$$

$$(A - \lambda I)x = 0$$

Since x is non zero, det(A-λI) = 0

With this characteristic equation, λ can be found.

With the eigenvalues, eigenvectors can be found.

Example

Given

$$A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\det \begin{bmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{bmatrix} = 0$$

$$(2-\lambda)(-6-\lambda) - (3)(3) = 0$$

$$\lambda = 3, -7$$

Example (continue)

$$(A - \lambda I)x = 0$$

$$\begin{bmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

For λ = 3,

$$\begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$-x_1 + 3x_2 = 0$$
$$3x_1 - 9x_2 = 0$$

Both give $x_1 = 3x_2$

So, eigenvector corresponding to λ = 3 is:

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Similarly, with λ = -7, the eigenvector is $[\ 1\ 3\ ]^T$

More about Eigenvector and Eigenvalue

- The determinant of a matrix is the multiple of its eigenvalues.
- If a matrix has eigenvalue 0,it is a singular matrix.
- eigenvalues of symmetric matrices are all real numbers.
- nonsymmetric matrices may have imaginary eigenvalues.

Row Echelon Form
And
Reduced Row Echelon Form

Row Echelon Form

- The following matrices are in row echelon form (REF):

$$\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$$

- The leading entries (■) may have any nonzero value;
- the starred entries (*) may have any value (including zero)

Reduced Row Echelon Form

- The following matrices are in reduced row echelon form (RREF):

$$\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$$

- The leading entries (■) have values 1
- the starred entries (*) may have any value (including zero)

Elementary Row Operations
And
Gaussian Elimination

Gaussian Elimination

- An algorithm for solving systems of linear equations
- Named after Carl Friedrich Gauss (1777–1855), who invented this method to solve a system of 17 linear equations.
- This method was introduced earlier in a Chinese book “The Nine Chapters on the Mathematical Art” around 250BC.
- Method: Use <u>elementary row operations</u> to bring an <u>augmented matrix</u> to <u>row echelon form.</u>

Augmented Matrix

- Given a system of linear equations as follow:

$$a_1x_1 + b_1x_2 + c_1x_3 = y_1$$
$$a_2x_1 + b_2x_2 + c_2x_3 = y_2$$
$$a_3x_1 + b_3x_2 + c_3x_3 = y_3$$

It can be written in a matrix form **Ax = y**, and we get an Augmented matrix

$$\left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & y_1 \\ a_2 & b_2 & c_2 & y_2 \\ a_3 & b_3 & c_3 & y_3 \end{array}\right]$$

Elementary row operations

- 3 operations:
  - Interchange 2 rows
  - Multiply a row with a non zero constant
  - Add a multiple of one row to another row
- The aim of these operations is to bring an augmented matrix to its REF or RREF.

Example

- Given system of linear equations:

$$\begin{cases} x + y + z = 6 \\ 2x + 3y + 4z = 16 \\ x - y - z = 0 \end{cases}$$

Augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 4 & 16 \\ 1 & -1 & -1 & 0 \end{array}\right]$$

Example

Row2: Subtract 2xRow 1 from Row2

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 3 & 4 & 16 \\ 1 & -1 & -1 & 0 \end{array}\right]$$

$$\xrightarrow{R_2 : R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 1 & -1 & -1 & 0 \end{array}\right]$$

Example

Row : Subtract Row1 from Row3

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 1 & -1 & -1 & 0 \end{array}\right]$$

$$\xrightarrow{R_3 : R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & -2 & -2 & -6 \end{array}\right]$$

Example

Row3: add 2xRow2 to Row3

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & -2 & -2 & -6 \end{array}\right]$$

$$\xrightarrow{R_3 : R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 2 & 2 \end{array}\right]$$

Example

From the matrix we get, we already can find the solution:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 2 & 2 \end{array}\right]$$

From the last row:

$$2z = 2 \quad \rightarrow \quad z = 1$$

We can solve for y if we substitute this result to row 2:

$$y + 2z = 4 \quad \rightarrow \quad y + 2(1) = 4 \quad \rightarrow \quad y = 2$$

Example

Alternatively, we can continue with elementary row operations until we find RREF:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 2 & 2 \end{array}\right]$$

$$R_2 \leftarrow R_2 - R_3$$

$$R_3 \leftarrow R_3/2$$

$$R_1 \leftarrow R_1 - R_2 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

(x, y, z) = (3, 2, 1)

Gradients And Hessians

Gradients

- Consider now the function $f : \mathbb{R}^n \to \mathbb{R}$, which is a real-valued function of *n* independent variables.
- We typically gather the variables into a vector *x*
- the gradient of *f* at *x is* $\nabla f(x)$.

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \qquad \nabla f(x) = \begin{bmatrix} \dfrac{\partial f}{\partial x_1} \\ \vdots \\ \dfrac{\partial f}{\partial x_n} \end{bmatrix}$$

Hessians

- The matrix of second partial derivatives of *f* is known as the Hessian

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Example

$$f(x, y) = 4x^3y^2 - \sin(x)$$

$$\nabla f(x, y) = \begin{bmatrix} 12x^2y^2 - \cos(x) \\ \\ 8x^3y \end{bmatrix}$$

$$\nabla f^2(x, y) = \begin{bmatrix} 24xy^2 + \sin(x) & 24x^2y \\ \\ 24x^2y & 8x^3 \end{bmatrix}$$

Example

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$f(\mathbf{x}) = 4x_1^3 x_2^2 - \sin(x_1)$$

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 12x_1^2 x_2^2 - \cos(x_1) \\ \\ 8x_1^3 x_2 \end{bmatrix}$$

$$\nabla f^2(\mathbf{x}) = \begin{bmatrix} 24x_1 x_2^2 + \sin(x_1) & 24x_1^2 x_2 \\ \\ 24x_1^2 x_2 & 8x_1^3 \end{bmatrix}$$

Attachment

Basic Properties and Formulas

If $f(x)$ and $g(x)$ are differentiable functions (the derivative exists), c and n are any real numbers,

1. $(c\,f)' = c\,f'(x)$
2. $(f \pm g)' = f'(x) \pm g'(x)$
3. $(f\,g)' = f'\,g + f\,g'$ – **Product Rule**
4. $\left(\dfrac{f}{g}\right)' = \dfrac{f'\,g - f\,g'}{g^2}$ – **Quotient Rule**
5. $\dfrac{d}{dx}(c) = 0$
6. $\dfrac{d}{dx}(x^n) = n\,x^{n-1}$ – **Power Rule**
7. $\dfrac{d}{dx}(f(g(x))) = f'(g(x))\,g'(x)$
   This is the **Chain Rule**

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Attachment

Common Derivatives

$$\frac{d}{dx}(x) = 1$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(a^x) = a^x \ln(a)$$

$$\frac{d}{dx}(\mathbf{e}^x) = \mathbf{e}^x$$

$$\frac{d}{dx}(\ln(x)) = \frac{1}{x}, \quad x > 0$$

$$\frac{d}{dx}(\ln|x|) = \frac{1}{x}, \quad x \neq 0$$

$$\frac{d}{dx}(\log_a(x)) = \frac{1}{x \ln a}, \quad x > 0$$

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Any questions?